

ELECTROSTATIC POTENTIAL and CAPACITANCE

1. ELECTROSTATIC POTENTIAL

The work done per unit positive test charge in bringing it from infinity to a point in an electric field without accelerating it, is called electrostatic potential at that point

$$V = \frac{W}{q_0}$$

Where,

W = work done (Joule)

q_0 = positive test charge (Coulomb)

→ SI Unit of Potential → Volt (V)

$$1 \text{ Volt} = \frac{1 \text{ Joule}}{1 \text{ Coulomb}}$$

Properties of Electrostatic Potential

- ★ It is a scalar quantity.
- ★ Its SI unit is Volt (V).
- ★ It is independent of path.
- ★ It is zero at infinity.
- ★ It can be positive, negative or zero.

Think!

Electric field is related to potential as

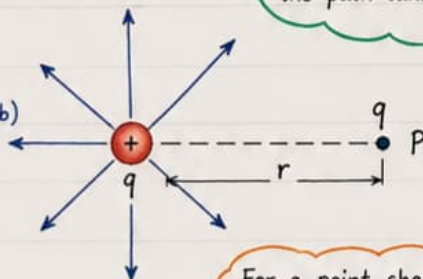
$$\vec{E} = -\nabla V$$

(Negative gradient of potential)

Key Idea

Potential is a scalar quantity.

It depends only on the position, not on the path taken.



For a point charge q ,

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

where,

ϵ_0 = permittivity of free space

Example 1:

Find the potential at a point 2 m away from a point charge of $3 \mu\text{C}$.

Solution:

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} = \frac{9 \times 10^9 \times 3 \times 10^{-6}}{2}$$

$$= 13.5 \times 10^3 \text{ V} = \boxed{13.5 \text{ kV}}$$



2. CAPACITANCE

The capacitance of a system is the charge stored per unit potential difference between the two conductors.

$$C = \frac{Q}{V}$$

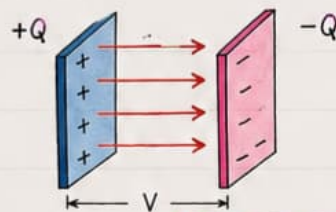
Where,

Q = charge on either conductor (C)

V = potential difference (V)

SI Unit of Capacitance → Farad (F)

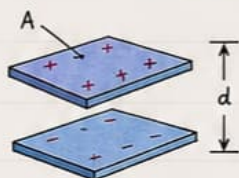
$$1 \text{ Farad} = \frac{1 \text{ Coulomb}}{1 \text{ Volt}}$$



Capacitance depends on the geometry and medium between the conductors.

Types of Capacitors

- Parallel Plate Capacitor
- Spherical Capacitor
- Cylindrical Capacitor



Parallel Plate Capacitor

$$C = \frac{\epsilon_0 A}{d}$$

where,

A = area of each plate

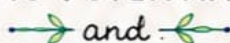
d = separation between plates

(In air/vacuum)

$$C = K \frac{\epsilon_0 A}{d}$$

(With dielectric)

Note: $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$



② Potential Due to a Point Charge

The electric potential at a point P, distance r from a point charge q is given by -

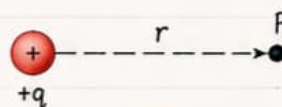
$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

where,

q = point charge

r = distance from charge

ϵ_0 = permittivity of free space



Potential is positive for $+q$ and negative for $-q$.

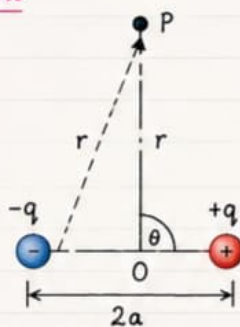
Potential due to Electric Dipole

At any point P on axial line

$$V_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2p \cos \theta}{r^2}$$

Along axial line, $\theta = 0^\circ$

$$V_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^2}$$



At any point P on equatorial line

$$V_{\text{equatorial}} = -\frac{1}{4\pi\epsilon_0} \frac{p}{r^2}$$

Along equatorial line, $\theta = 90^\circ$

$$V_{\text{equatorial}} = -\frac{1}{4\pi\epsilon_0} \frac{p}{r^2}$$

Example 2:

Find the potential at a point 5 cm away from a charge of $2 \mu\text{C}$.

Solution:

$$\begin{aligned} V &= \frac{1}{4\pi\epsilon_0} \frac{q}{r} \\ &= 9 \times 10^9 \times \frac{2 \times 10^{-6}}{5 \times 10^{-2}} \\ &= 3.6 \times 10^5 \text{ V} \end{aligned}$$

$$p = q(2a)$$

p = dipole moment
(unit : C m)

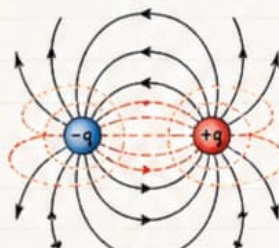
Equipotential Surfaces

- An equipotential surface is a surface on which electric potential is same at every point.
- No work is done in moving a charge on an equipotential surface.
- Electric field is always perpendicular to equipotential surface.

For a positive point charge



For an electric dipole



----- Equipotential surface

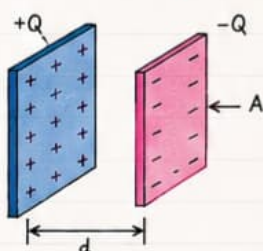
→ Electric field line

Points to Remember

- ★ Closer the equipotential surfaces, stronger is the electric field.
- ★ No two equipotential surfaces can intersect.
- ★ Conductors in electrostatic equilibrium are equipotential.

③ CAPACITOR - more on page 1

② Parallel Plate Capacitor



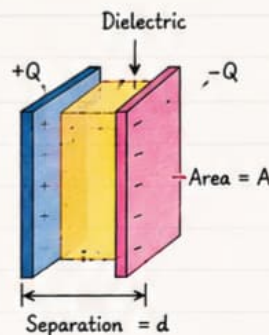
Capacitance,

$$C = \frac{\epsilon_0 A}{d}$$

→ If a dielectric is inserted between plates,

$$C = K \frac{\epsilon_0 A}{d} = KC_0$$

where K = dielectric constant ($K > 1$)



Energy stored in a capacitor

$$U = \frac{1}{2} CV^2$$

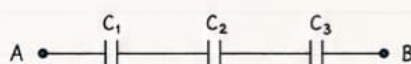
where,

C = capacitance

V = potential difference

④ Combination of Capacitors

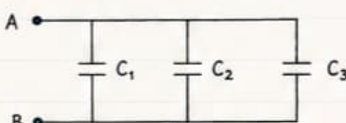
(i) Capacitors in Series



$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

Charge on each capacitor is same.

(ii) Capacitors in Parallel



$$C_{\text{eq}} = C_1 + C_2 + C_3 + \dots$$

Potential difference across each capacitor is same.

Example 3:

Two capacitors of $2 \mu\text{F}$ and $3 \mu\text{F}$ are connected in series across a 6 V battery. Find (i) equivalent capacitance (ii) charge on each.

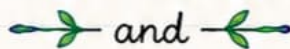
Solution:

$$(i) \frac{1}{C_{\text{eq}}} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

$$C_{\text{eq}} = \frac{6}{5} = 1.2 \mu\text{F}$$

$$(ii) \text{ Charge } Q = C_{\text{eq}} V = 1.2 \times 6 = 7.2 \mu\text{C} \text{ (same on each)}$$





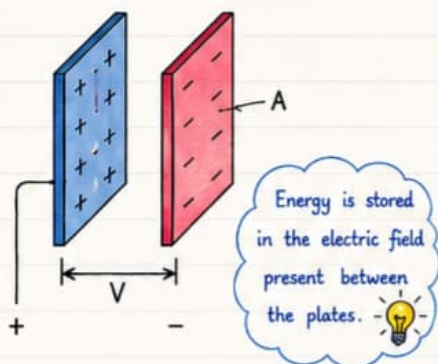
5 Work Stored in Capacitor

→ Work done in charging a capacitor is stored as potential energy in it.

$$U = \frac{1}{2} CV^2$$

$$= \frac{Q^2}{2C}$$

$$= \frac{1}{2} QV$$



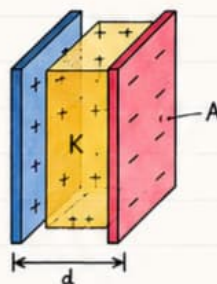
6 Effect of Dielectric on Capacitance

★ When a dielectric slab (dielectric constant = K) is inserted between the plates:

$$C' = KC$$

Where, C = capacitance without dielectric

C' = capacitance with dielectric ($K > 1$)

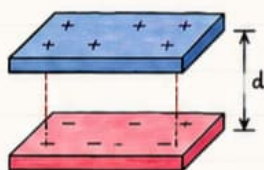


Why Capacitance increases?

★ Dielectric reduces the electric field between plates for the same potential difference. Hence, more charge can be stored → Capacitance increases.

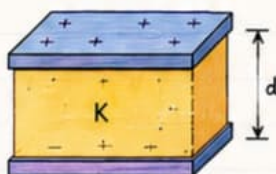
7 Capacitors in Different Situations

(i) Parallel Plate Capacitor (air/vacuum)



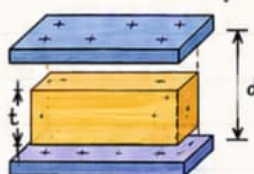
$$C = \frac{\epsilon_0 A}{d}$$

(ii) With Dielectric Slab (filling entire space)



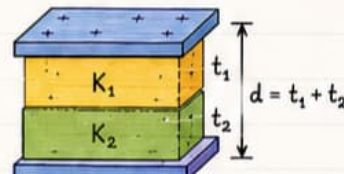
$$C = K \frac{\epsilon_0 A}{d}$$

(iii) With Dielectric Slab (partial filling)



$$C = \frac{\epsilon_0 A}{d - t + \frac{t}{K}}$$

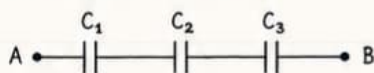
(iv) Two Dielectrics (in series)



$$C = \frac{\epsilon_0 A}{\frac{t_1}{K_1} + \frac{t_2}{K_2}}$$

8 Combinations of Capacitors

(i) Capacitors in Series



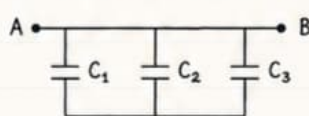
$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

Same charge (Q) on each capacitor.

$$V = V_1 + V_2 + V_3 + \dots$$

$$\frac{V_1}{V_2} = \frac{C_2}{C_1}$$

(ii) Capacitors in Parallel



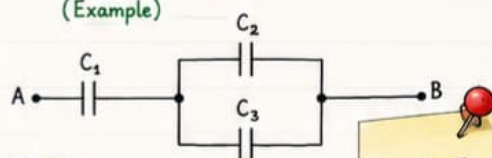
$$C_{eq} = C_1 + C_2 + C_3 + \dots$$

Same potential difference (V) across each capacitor.

$$Q = Q_1 + Q_2 + Q_3 + \dots$$

$$\frac{Q_1}{Q_2} = \frac{C_1}{C_2}$$

(iii) Series-Parallel Combination (Example)



Solution :

C_2 and C_3 are in parallel

$$\rightarrow C_{23} = C_2 + C_3$$

Now C_1 is in series with C_{23}

$$\rightarrow \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_{23}}$$

Quicks Tip

See connection carefully first, then apply series or parallel formula. ★

Points to Remember

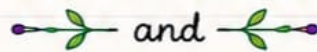
- ★ Potential is scalar, depends on position.
- ★ Capacitance is charge stored per unit potential difference.
- ★ Capacitance depends on geometry and dielectric medium.
- ★ Energy stored in a capacitor is due to electric field between plates.
- ★ Dielectric increases capacitance.

Important Formulae

- $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$ (point charge)
- $V = \frac{1}{4\pi\epsilon_0} \frac{2p \cos \theta}{r^2}$ (dipole axial)
- $V = \frac{1}{4\pi\epsilon_0} \frac{p}{r^2}$ (dipole equatorial)
- $C = \frac{\epsilon_0 A}{d}$ (parallel plate)
- $U = \frac{1}{2} CV^2$ (energy stored)

Units & Dimensions

- Potential (V) → Unit : Volt (V)
Dimension : $[M^1 L^2 T^{-3} A^{-1}]$
- Capacitance (C) → Unit : Farad (F)
Dimension : $[M^{-1} L^{-2} T^4 A^2]$
- Energy (U) → Unit : Joule (J)
Dimension : $[ML^2 T^{-2}]$



9 Energy Density in an Electric Field

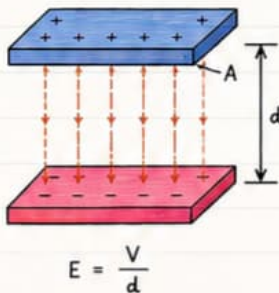
→ The energy stored in a capacitor may be considered to be present in the space between the plates in the form of electric field.

$$u = \frac{U}{\text{Volume}} = \text{energy per unit volume}$$

For parallel plate capacitor,
Volume = Ad

$$u = \frac{\frac{1}{2} CV^2}{Ad} = \frac{1}{2} \epsilon_0 E^2$$

where, $E = \frac{V}{d}$



★ Meaning :

This tells how much energy is stored in unit volume of space between the plates.



Think !

More the electric field (E) between the plates, more will be the energy density stored.

10 Important Points ★

- ★ Potential is zero at infinity.
- ★ Potential due to multiple charges is algebraic sum of potentials due to individual charges.
- ★ Capacitance of a system depends on geometry and dielectric medium.
- ★ In series combination, equivalent capacitance is less than the smallest capacitance.
- ★ In parallel combination, equivalent capacitance is greater than the largest capacitance.
- ★ Energy stored in a capacitor is associated with electric field in the space between plates.

Commonly Used Relations

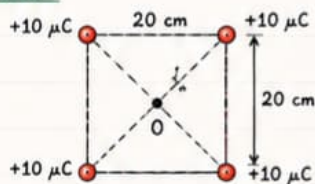
- $V = \frac{W}{q_0}$
- $C = \frac{Q}{V}$
- $U = \frac{1}{2} CV^2 = \frac{Q^2}{2C} = \frac{1}{2} QV$
- $E = \frac{V}{d}$
- $u = \frac{1}{2} \epsilon_0 E^2$

Units at a glance

- Potential (V) → Volt (V)
- Capacitance (C) → Farad (F)
- Energy (U) → Joule (J)
- Charge (Q) → Coulomb (C)

- ① Find the potential at the centre of a square of side 20 cm if $10 \mu C$ charge is placed at each corner.

Solution :



Distance of centre from each corner,
 $r = \frac{a}{\sqrt{2}} = \frac{20}{\sqrt{2}} \text{ cm} = 10\sqrt{2} \text{ cm}$

Potential at centre due to one charge,
 $V_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$

Total potential, $V = 4 V_1$

$$= \frac{4}{4\pi\epsilon_0} \frac{q}{r}$$

$$= \frac{9 \times 10^9 \times 4 \times 10^{-5}}{10\sqrt{2} \times 10^{-2}}$$

$$= 2.55 \times 10^5 \text{ V}$$

- ② A $4 \mu F$ capacitor is charged to 100 V.
Find (i) charge stored (ii) energy stored.

Solution :

(i) Charge stored,

$$Q = CV = 4 \times 10^{-6} \times 100$$

$$Q = 4 \times 10^{-4} \text{ C (400 } \mu C)$$

(ii) Energy stored,

$$U = \frac{1}{2} CV^2$$

$$= \frac{1}{2} \times 4 \times 10^{-6} \times (100)^2$$

$$U = 2 \times 10^{-2} \text{ J (0.02 J)}$$

Quick Revision

- ① Potential is scalar.
- ② Potential at infinity = 0
- ③ Capacitance depends on geometry.
- ④ Dielectric increases capacitance.
- ⑤ Energy stored = $\frac{1}{2} CV^2$
- ⑥ Electric field stores energy. ★

- ③ Three capacitors of $2 \mu F$, $3 \mu F$ and $6 \mu F$ are connected in parallel. Find equivalent capacitance and charge on each capacitor if the combination is connected to a 12 V battery.

Solution :

(i) Equivalent capacitance,

$$C_{eq} = C_1 + C_2 + C_3$$

$$= 2 + 3 + 6 = 11 \mu F$$

$$C_{eq} = 11 \mu F$$

(ii) Charge on each capacitor,

$V = 12 \text{ V}$ (same across each)

$$Q_1 = C_1 V = 2 \times 12 = 24 \mu C$$

$$Q_2 = C_2 V = 3 \times 12 = 36 \mu C$$

$$Q_3 = C_3 V = 6 \times 12 = 72 \mu C$$

Total charge,

$$Q = C_{eq} V = 11 \times 12 = 132 \mu C$$

$$Q = 132 \mu C$$

Memory Tip



High Potential → High energy
Large separation → Low capacitance
Small separation → High capacitance
Dielectric → Capacitance increase

Real Life Application

Capacitors are used in:-

- Flash lights, Camera, Defibrillators
- Filters in power supplies
- Energy storage systems



Exam Tip

- ✓ Always check the sign of charge for potential.
- ✓ Remember : V is scalar, $\vec{E}$ is vector.
- ✓ Draw diagram first in every numerical.
- ✓ Use correct formula for correct situation.

